EE 5239 Course Project Proposal

Optimization for Multimodal Sensor Fusion

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Fall 2026

Project Description

The goal of this project is to investigate the use of optimization methods for multimodal sensor fusion. Many engineering systems use multiple sensors to estimate a quantity or state that cannot be measured accurately using a single sensor. The performance of these systems depends on how information from the available sensors is combined. This project will investigate how optimization can be used to determine an effective method of combining measurements from multiple sensors.

A publicly available dataset containing simultaneous measurements from multiple sensors will be selected for the project. Preference will be given to physiological sensor data, such as measurements related to cardiovascular, respiratory, or activity signals. The selected data will be processed and divided into appropriate training and testing datasets.

A sensor fusion model will then be developed to estimate a target quantity using the available sensor measurements. The parameters of this model will be determined using optimization. Different optimization methods will be implemented and compared to investigate their convergence behavior, computational requirements, and resulting estimation performance. The specific model and optimization algorithms will be selected after reviewing the characteristics of the chosen dataset and relevant literature.

The project will also investigate the importance of the individual sensors. Models using the complete sensor set will be compared with models using reduced sensor sets to determine whether similar estimation performance can be achieved with fewer measurements. If appropriate for the selected model, regularization or other optimization-based approaches to sensor selection will also be investigated.

The project will be implemented primarily in Python using standard numerical and data-analysis libraries. The expected computational requirements should be suitable for a personal computer. Results will be evaluated using appropriate estimation error metrics as well as measures of optimization performance, such as convergence behavior and computational time.

The expected outcome of the project is a comparison of optimization approaches for multimodal sensor fusion and an analysis of the tradeoff between estimation performance, computational requirements, and the number of sensors used. Additional model complexity or alternative optimization formulations may be investigated if time permits.